Migraine is a Risk Factor for Ischemic Stroke

Timothy T. Houle, PhD

HARVARD MEDICAL SCHOOL

MASSACHUSETTS GENERAL HOSPITAL

Disclosures

- No financial relationships with industry
- Statistical Editor (*Anesthesiology*)
- Statistical Consultant (*Headache*)
- NIH grants:
  - NS065257
  - GM113852

Migraine & Stroke

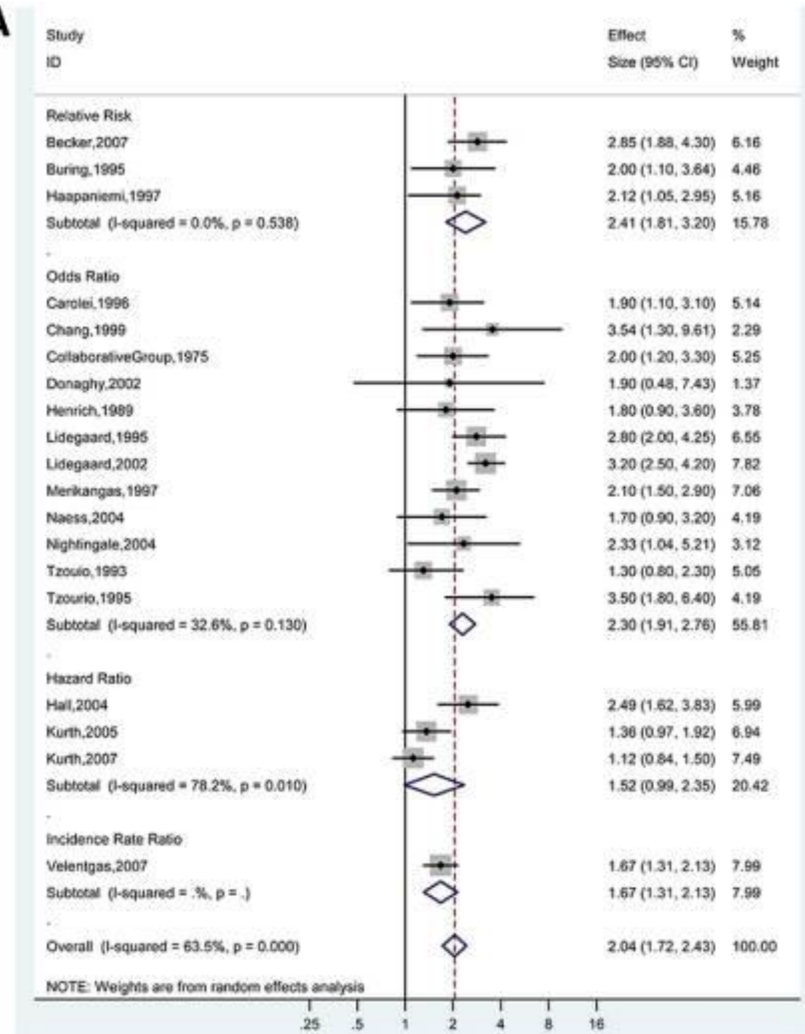
- Ischemic Stroke
- Hemorrhagic stroke
- Migrainous infarction
  - During migraine attack with aura (aura > 60 min)
  - 0.2% to 0.5% of all ischemic strokes
- Cerebral Autosomal Dominant Arteriopathy with Subcortical Infarcts and Leukoencephalopathy (CADASIL)
- Mitochondrial Encephalopathy, Lactic Acidosis and Stroke-like episodes (MELAS)

Epidemiology: Migraine vs no migraine

- 35 studies (N = 622,381)
  - RR: 2.41
    (95%CI: 1.81-3.20)
  - HR: 1.52
    (95%CI: 0.99-2.35)

  Overall:
  RR 2.04 (95% CI, 1.72-2.43)

A



Migraine Headache and Ischemic Stroke Risk: An Updated Meta-analysis (2010)
Spector, Kahn, Jones, Jayakumar, Dalal, Nazarian

Epidemiology: Migraine subgroups

<u>Pooled relative risks (95%CI)</u>

- Any migraine: 1.73 (1.31 to 2.29)
- migraine w/ aura: 2.16 (1.53 to 3.03)
- migraine w/out aura: 1.23 (0.90 to 1.69)

- Women: 2.08 (1.13 to 3.84)
- Men: 1.37 (0.89 to 2.11)

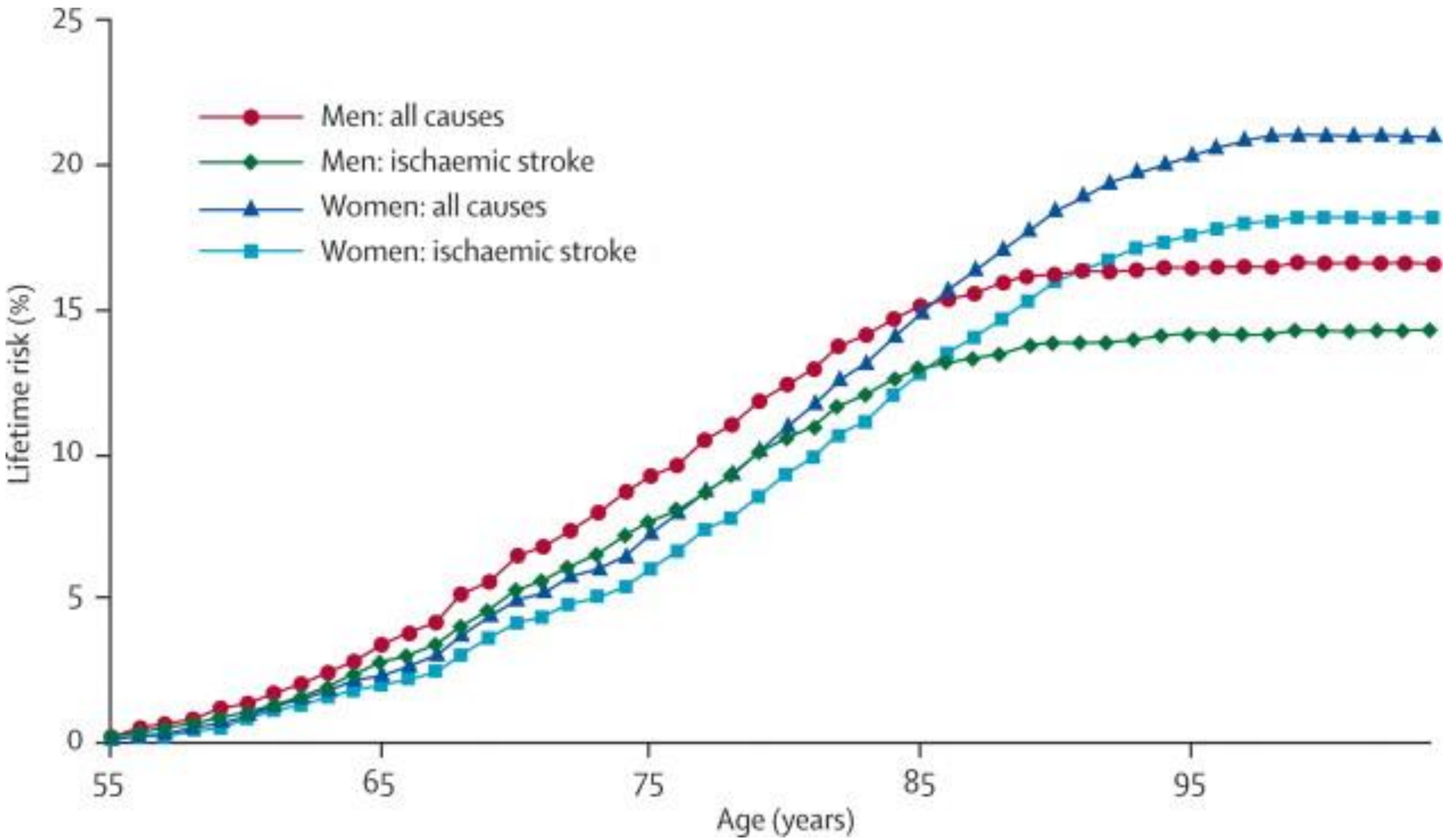
Migraine and cardiovascular disease: systematic review and meta-analysis (2009)
Schurks, Rist, Bigal, Buring, Lipton, Kurth

“Classic” Stroke Risk Factors for Ischemic Stroke in Migraine

- Migraine with aura
- Women
- Age <45 years
- Oral contraceptive use*
- Smoking

*Never fear, Anne Calhoun is here

Lifetime Risk of Stroke



Lifetime risk of stroke and dementia: current concepts, and estimates from the Framingham Study. (2007) Seshadri et al.

Epidemiology: Migraine-related stroke

For 100,000 women <= 35 year-old

	No oral contraceptive	Oral contraceptive
No migraine	1.3	5
Migraine w/out aura	4	14
Migraine w/ aura	8	28

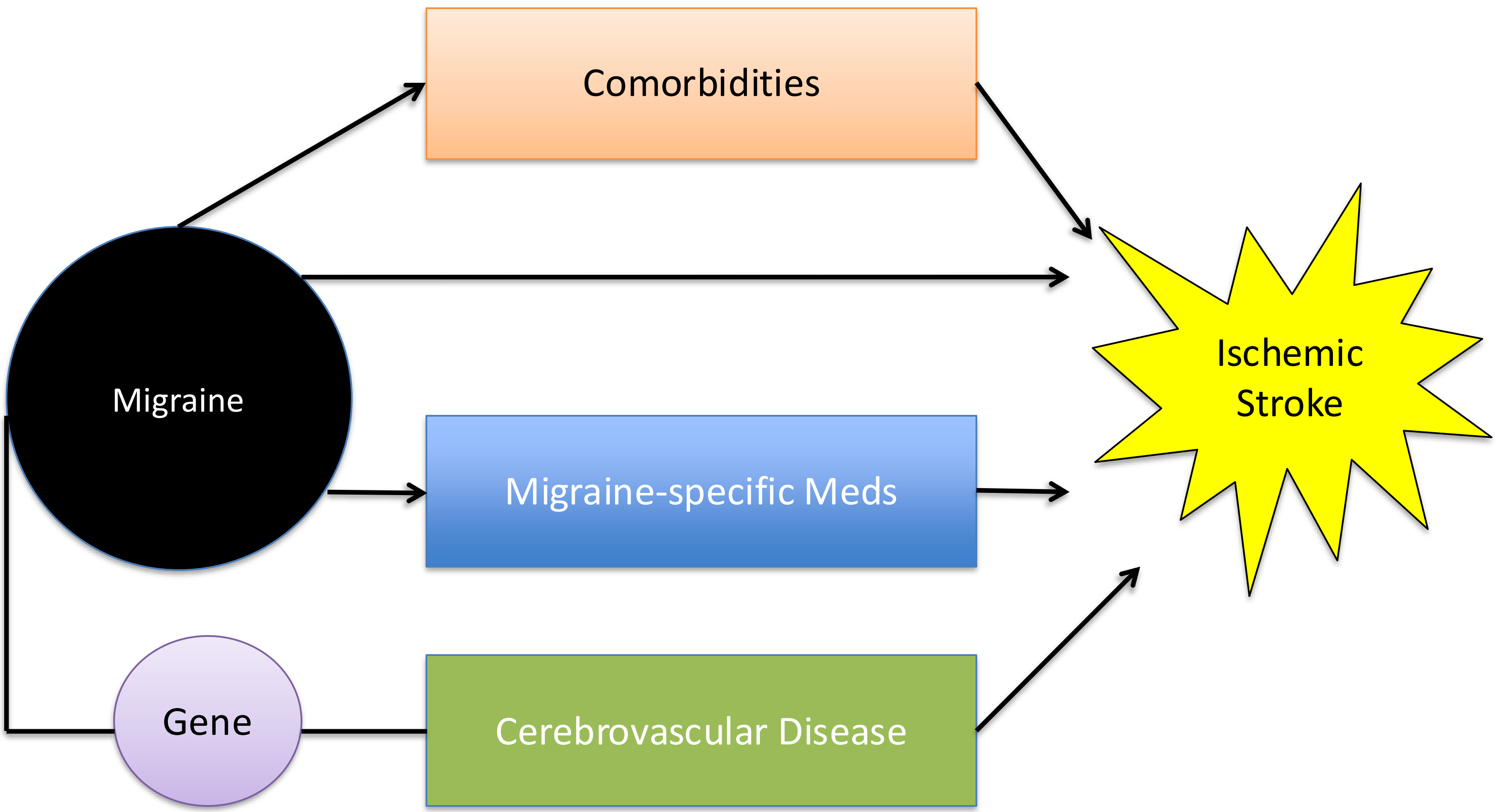
Source: Migraine Trust

8 per 100,000 die in automobile accident

175 per 100,000 stroke incidence in general population

Global Burden of Disease Study 2013

Proposed Mechanisms of Migraine-Stroke Relationship



Possible Mechanisms

- Genetic predisposition
  - MTHFR C677T, ACE-DD polymorphism
  - rs7698623 in *MEPE,* rs4975709 in *IRX4*
- Endothelial dysfunction
- Coagulation dysfunction
  - Decreased resistance to activated protein C
- Arterial dissection
  - Elevated serum elastase activity
- Patent foramen ovale (PFO)

The Migraine–Stroke Connection (2016). Lee et al

Migraine and risk of perioperative ischemic stroke and hospital readmission: hospital based registry study

Fanny P Timm *medical student*[1], Timothy T Houle *associate professor*[1], Stephanie D Grabitz *medical student*[1], Anne-Louise Lihn *medical student*[1 2], Janne B Stokholm *medical student*[1 2], Katharina Eikermann-Haerter *assistant professor*[3], Ala Nozari *associate professor*[1], Tobias Kurth *professor*[4], Matthias Eikermann *clinical director*[1]

[1]Department of Anesthesia, Critical Care, and Pain Medicine, Massachusetts General Hospital and Harvard Medical School, 55 Fruit Street, Boston, MA 02114, USA; [2]Department of Anesthesiology, University of Copenhagen, Herlev Hospital, Copenhagen, Denmark; [3]Department of Radiology, Massachusetts General Hospital and Harvard Medical School, Boston, MA, USA; [4]Institute of Public Health, Charité - Universitätsmedizin Berlin, Germany

thebmj 2017

Ischemic Stroke in Perioperative Period

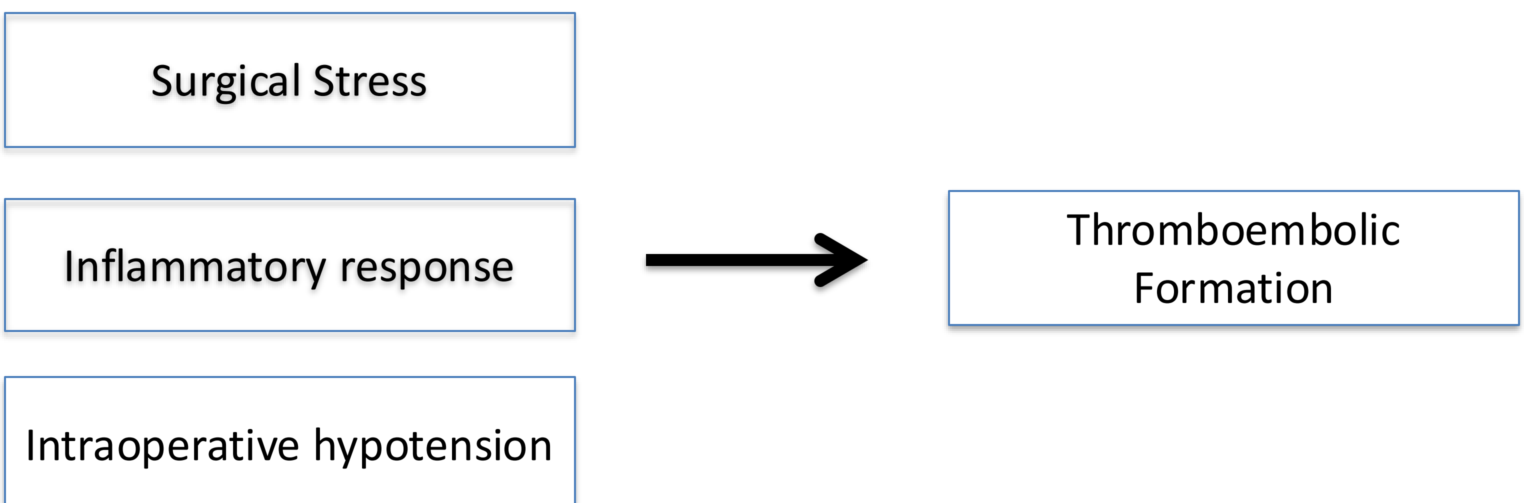
- 100 strokes per 100,000 low risk surgeries[1]
- 600 to 7,400 per 100,000 after major cardiac and vascular surgery[2]
- Stroke risk is radically increased compared to background risk

[1] Mashour GA, Shanks AM, Kheterpal S. Perioperative stroke and associated mortality after noncardiac, nonneurologic surgery. Anesthesiology. 2011 Jun 1;114(6):1289-96.

[2] Bucerius J, Gummert JF, Borger MA, Walther T, Doll N, Onnasch JF, Metz S, Falk V, Mohr FW. Stroke after cardiac surgery: a risk factor analysis of 16,184 consecutive adult patients. The Annals of thoracic surgery. 2003 Feb 1;75(2):472-8.

Mechanisms for Stroke Risk Near Surgery

- Indication for surgery (e.g., vascular comorbidity)



Hypothesis

- Given the hyper-risk for perioperative ischemic stroke, studying this risk period could yield unique insights into the migraine-stroke connection

- We hypothesized that individuals with migraine would have an elevated risk of stroke within 30 days of surgery

Study Design

- Retrospective cohort study

- All surgeries conducted at Massachusetts General Hospital (MGH)
  - 2007 to 2014
  - Patients who were extubated

- Data sources:
  - Anesthesia information management system
  - Research Patient Data Registry

Exposure & Outcomes

- Primary Exposure: Migraine identified using ICD-9 codes
  - Patients without a relevant code, coded as NOT having migraine
  - Bias: this will almost certainly undercount migraine

- Primary Outcome: Ischemic stroke occurring within 30 days of surgery
  - ICD-9 codes ischemic stroke
  - Hand-checked all flagged records (MRI, CT, consultant reports)

Statistical Analyses

- Multivariable logistic regression adjusting for stroke risk factors
  - Migraine vs. no-Migraine
  - Migraine w/aura, Migraine wo/aura, no-Migraine

- Multinomial regression for stroke location
- Sensitivity models
  - history of transient ischemic attack (TIA)
  - Background stroke risk
  - Multiple imputation for missing data

Confounder Model

- Sex
- Age
- BMI
- ASA
- Emergent
- Inpatient Surgery
- High Risk Surgery
- Charlson Comorbidity Index
- Antiplatlet drugs
- Beta-blockers (4 weeks before surgery)
- Coronary artery disease
- Dyslipidemia
- Diabetes
- Hypertension
- Atrial fibrillation
- Right-to-left shunt (PFO)
- Duration of surgery
- Intraoperative hypotension duration
- Intraoperative vasopressor dose
- Intraoperative fluid volume
- Blood transfusion
- Work relative value units

Results

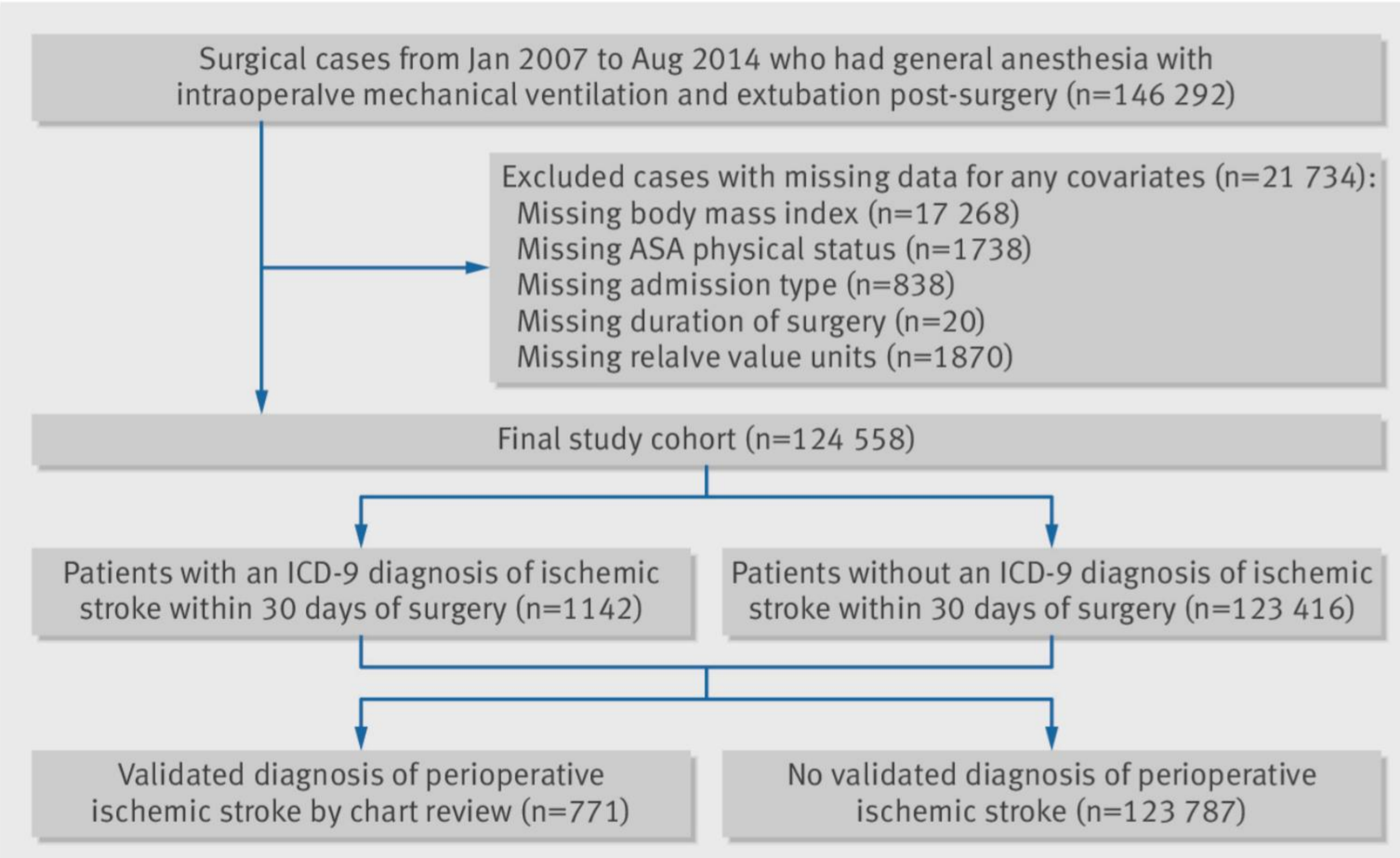


Fig 1 Flow chart of study. ASA=American Society of Anesthesiologists

Migraine Classification

- 10,179 (8.2%) individuals with migraine
  - 8,901 Migraine without aura
  - 1278 Migraine with aura

- Lower than expected, indicates bias

Characteristics	Total study population (n=124 558)	No migraine (n=114 379)	Migraine diagnosis (n=10 179)		
			Any migraine (n=10 179)	Migraine without aura (n=8901)	Migraine with aura (n=1278)
Mean (SD) age, years	52.6 (18.3)	53.1 (18.4)	47.4 (15.9)	47.7 (15.9)	45.1 (15.2)
Female sex	67 943 (54.6)	59 800 (52.3)	8143 (80.0)	7105 (79.8)	1038 (81.2)
Mean (SD) body mass index	28.3 (7.3)	28.3 (7.2)	28.9 (8.1)	29.0 (8.1)	28.3 (7.6)
[Median (IQR)?]ASA physical status classification	2 (2-3)	2 (2-3)	2 (2-3)	2 (2-3)	2 (2-3)
Emergency status	5469 (4.4)	5085 (4.5)	384 (3.8)	340 (3.8)	44 (3.4)
Inpatient	94 929 (76.2)	87 299 (76.3)	7630 (75.0)	6721 (75.5)	909 (71.1)
Median (IQR) Charlson Comorbidity Index	1 (0-3)	1 (0-3)	1 (0-2)	1 (0-2)	1 (0-3)
Any prescription of antiplatelet drug within 4 weeks before surgery	14 054 (11.3)	13 214 (11.6)	840 (8.3)	730 (8.2)	110 (8.6)
Any prescription of β blockers within 4 weeks before surgery	16 307 (13.1)	15 227 (13.3)	1080 (10.6)	926 (10.4)	154 (12.1)
Stroke risk factors:					
Diabetes	8971 (7.2)	8509 (7.4)	462 (4.5)	405 (4.6)	55 (4.3)
Hypertension	51 601 (41.4)	47 764 (41.8)	3837 (37.7)	3353 (37.7)	484 (37.9)
Dyslipidemia	39 115 (31.4)	35 897 (31.4)	3218 (31.6)	2795 (31.4)	423 (33.1)
Coronary artery disease	8662 (7.0)	8077 (7.1)	585 (5.8)	494 (5.5)	91 (7.1)
Atrial fibrillation	8088 (6.5)	7607 (6.7)	481 (4.7)	397 (4.5)	84 (6.6)
Possible right-to-left shunt	1530 (1.2)	1325 (1.2)	205 (2.0)	155 (1.7)	50 (3.9)
History of transient ischemic attack	1744 (1.4)	1419 (1.2)	325 (3.2)	227 (2.6)	98 (7.7)
Intraoperative characteristics:					
Median (IQR) duration of procedure, hours	2.4 (1.5-3.8)	2.5 (1.5-3.8)	2.4 (1.5-3.7)	2.4 (1.5-3.7)	2.3 (1.4-3.5)
Median (IQR) intraoperative hypotensive minutes <MAP 55 mm Hg	0 (0-2)	0 (0-2)	0 (0-2)	0 (0-2)	0 (0-2)
Median (IQR) total intraoperative norepinephrine equivalent dose, mg	0 (0-0.2)	0 (0-0.2)	0 (0-0.1)	0 (0-0.1)	0 (0-0.1)

Table 2| **Incidences and univariable and multivariable adjusted odds ratios for study outcomes and strata of stroke location according to migraine status. Values are numbers (percentages) unless stated otherwise**

Outcome	Total (n=124 558)	No migraine (n=114 379)	Migraine (n=10 179)	Odds ratio (95% CI)	
				Unadjusted	Adjusted
Perioperative ischemic stroke	771 (0.6)	682 (0.6)	89 (0.9)	1.47 (1.18 to 1.84); P=0.001	1.75 (1.39 to 2.21)*; P<0.001
Stroke subtypes according to Oxfordshire Community Stroke Project classification†:					
Total anterior stroke	223 (0.2)	205 (0.2)	18 (0.2)	0.99 (0.61 to 1.6); P=0.97	1.21 (0.74 to 1.97)‡; P=0.44
Partial anterior stroke	102 (0.1)	89 (0.1)	13 (0.1)	1.65 (0.92 to 2.95); P=0.09	1.95 (1.08 to 3.52)‡; P=0.03
Posterior circulation stroke	181 (0.1)	158 (0.1)	23 (0.2)	1.64 (1.06 to 2.54); P=0.03	1.96 (1.26 to 3.05) ‡; P=0.003
Lacunar stroke	30 (0)	28 (0)	2 (0)	0.80 (0.19 to 3.38); P=0.77	1.13 (0.27 to 4.79) ‡; P=0.87
Unclassifiable	235 (0.2)	202 (0.2)	33 (0.3)	1.84 (1.27 to 2.66); P=0.001	2.33 (1.61 to 3.39) ‡; P<0.001
Median (interquartile range) days until stroke	3 (1-11)	3 (1-10)	5.5 (0-14.5)		
Post-discharge stroke	195 (0.2)	164 (0.1)	31 (0.3)	1.76 (1.08 to 2.85); P=0.02	1.94 (1.18 to 3.20) ‡; P=0.009
Hospital readmission within 30 days	10 088 (8.1)	9160 (8.0)	928 (9.1)	1.16 (1.08 to 1.24); P<0.001	1.31 (1.22 to 1.41)*; P<0.001

*Adjusted for age; sex; body mass index; American Society of Anesthesiologists physical status classification; emergent *v* non-emergent surgery; admission type (inpatient); Charlson Comorbidity Index; history of diabetes, hypertension, atrial fibrillation, dyslipidemia, possible right-to-left-shunt, coronary artery disease; any prescription of antiplatelet drug or β blockers within 4 weeks before surgery; duration of surgery; total intraoperative fluid volume; total intraoperative vasopressor dose (norepinephrine equivalent); intraoperative packed red blood cell units received; work relative value units.

†Associations between stroke subtypes (reference: no stroke) and migraine were analyzed using a multinomial logistic regression model.

‡Adjusted for age.

Multiple Imputation, Sensitivity Analyses, TIA, PFO

Comparison of Risks

- Schurks et al (2009)

<u>Any migraine</u>:
1.73 (1.31 to 2.29)

<u>Migraine w/ aura</u>:
2.16 (1.53 to 3.03)

<u>Migraine w/out aura</u>:
1.23 (0.90 to 1.69)

- Current Study (2017)

<u>Any migraine</u>:
1.75 (1.39 to 2.21)

<u>Migraine w/ aura</u>:
2.61 (1.59 to 4.29)

<u>Migraine w/out aura</u>:
1.62 (1.26 to 2.09)

Migraine may be a Unique Risk Factor

- Created a 'background' risk score of risk factors excluding migraine
  - This risk score performed well in discriminating stroke risk (AUC: 0.85)

Low Risk	Intermediate	High
3.52 (1.28 to 9.7)	1.62 (0.83 to 3.16)	1.63 (1.28 to 2.08)

Relative risk of migraine by background risk groups

Absolute Risks

- 240 per 100,000 patients experience ischemic stroke with 30 days after surgery

- 430 per 100,000 for any migraine diagnosis
  - 390 per 100,000 migraine without aura
  - 630 per 100,000 migraine with aura

Our Next Steps

- PFO is a unique risk factor for ischemic stroke during surgery
  - 30 days: Ng et al JAMA 2018
  - 1 year: Friedrich et al European Heart Journal 2018

- Individuals with migraine are readmitted to hospital more frequently due to pai
  - Platzbecker et al., Cephalalgia 2018

- A stroke prediction model
  - Under review

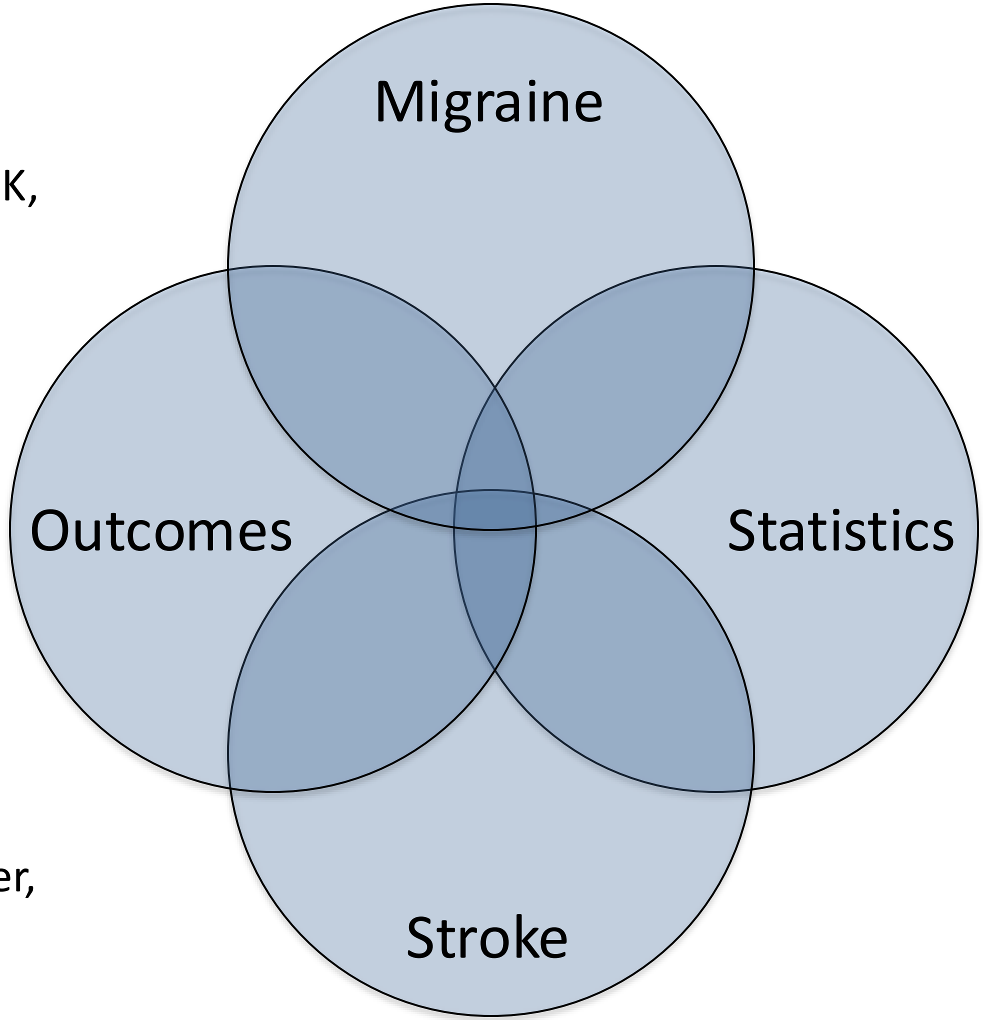
Conclusions

- Migraine is a risk factor for ischemic stroke
  - As a background risk (women, smoking, OC)
  - During surgery
- Migraine with aura is of particular concern
- The perioperative period represents an opportunity to study the migraine-risk connection
- Research should focus on understanding the mechanisms underlying this risk

Collaborators

MASSACHUSETTS GENERAL HOSPITAL

Stephanie Grabitz
Lihn AL,
Stokholm JB,
Eikermann-Haerter K,
Nozari A,
Tobias Kurth
Pauline Ng,
Ng AK,



Sara Burns
Herisson F,
Med C,
Rudolph MI,
Flora Scheffenbichler,
Sabine Friedrich
Deepak Bhatt
Dana Turner
Bhala Subramaniam

Matthias Eikermann

Fanny Timm

Beth Israel Deaconess Medical Center